Himanshu Sharma

B.S. Physics (3rd Year) | Himanshu Sharma | himans23@iitk.ac.in | +91-9999633455 | GitHub

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023 - Present	B.S. (Physics)	Indian Institute of Technology, Kanpur	**8.0/10**
2023	CBSE(XII)	Amity International School, Saket	**98.0%**
2021	CBSE(X)	Amity International School, Saket	**98.2%**

Scholastic Achievements

- Secured **All India Rank of 4506** in JEE Advanced 2023 among the **1,20,000+ candidates** across India *(2023)*
- Qualified **Pre Regional Mathematics Olympiad**, ranked among **top 300** in Delhi region *(2019)*

Relevant Coursework (*) represents ongoing

- **Physics:** Quantum Physics (A), Quantum Computing and Communication (A), Nanophotonics (A), Quantum Mechanics II (*), Quantum Processes in Low Dimensional Semiconductors, Quantum Optics (*), Function Spaces for Formalism of Quantum Mechanics (*), Mathematical Methods for Physics, Classical Electrodynamics (*), Statistical Mechanics (*)
- **Labs:** Modern Physics Lab (A), Optics Lab, Electronics Lab

Technical Skills

- **Programming:** Python, MATLAB, C, C++
- **Computational Physics & Toolkits:** QuTiP, Qiskit, COMSOL Multiphysics, OriginPro
- **Machine Learning and Data Analysis:** Tensorflow, Keras, PyTorch, Transformerlens, OpenCV
- **Experimental Skills:** Crystal Growth, Chemical Vapor Transport, Scanning Electron Microscopy, X-ray Diffraction, Energy-Dispersive X-ray Spectroscopy

Summer School

IBM Quantum Global Summer School 2025
(July'25)

- Developed a theoretical and hands-on understanding of Quantum Computing, **Simulation**, **Advanced Algorithms**, **Architectures**, **Hardware**, **Noise Characterization**, **Error Correction**, **Benchmarking**.
- Engineered the mathematical structure of **Quantum Error Correction** (QEC) codes by constructing the stabilizer parity check matrices for locally-connected **Toric code** and **non-local Gross** (QLDPC) code.
- Characterized, mitigated computational errors by implementing **Zero Noise Extrapolation** with global and local **circuit folding**, demonstrating practical approach to managing decoherence in quantum systems.
- Determined ground state energy of molecular Hamiltonian (N_2) using hybrid **Sample-based Quantum Diagonalization** algorithm, employing physically-motivated **LUCJ ansatz** to prepare the quantum state.
- Developed a post-processing workflow to correct noisy experimental data by implementing a **self-consistent configuration recovery** technique that enforces physical symmetries.

Selected Research Projects

High-Q SiN Microring Resonator for Diamond NV Center Purcell Enhancement | *Dr Rituraj, IIT Kanpur*
Course Project, Nanophotonics *(Sept'25 - Nov'25)*

- Designed and simulated a **SiN-on-SiO$_2$ microring resonator** to enhance spontaneous emission of diamond NV centers at the **637 nm zero-phonon line**, targeting cavity-QED based spin–photon interfaces.
- Implemented a **2D axisymmetric eigenfrequency FEM model in COMSOL** with PML boundaries to compute whispering-gallery modes and quantify radiative loss.
- Performed **mesh-convergence analysis** and optimized ring geometry (radius 19.5–20.5 μm, 250 $\times$ 500 nm waveguide) to spectrally align the fundamental WGM ($m = 354$) with the NV ZPL, extracting Q and mode volume.
- Analyzed the Q–V **trade-off** and benchmarked against prior literature, revealing confinement-induced loss mechanisms intrinsic to microring architectures.
- Achieved $Q \approx 1.7 \times 10^5$ with **Purcell enhancement** $F_p \approx 3.9$, demonstrating fundamental emission-rate limits of microrings and motivating **photonic crystal cavities** for high-efficiency NV quantum interfaces.

Quantum Optics Simulations, Cavity-QED Research
Dr Rituraj, IIT Kanpur *(Nov'25 - Ongoing)*

- Developed a unified computational and theoretical framework to study **light–matter interaction** in confined quantum systems and photonic quantum computing primitives.
- Analyzed the validity and breakdown of the **Rotating Wave Approximation (RWA)** in driven, time-dependent regimes.
- Compared the full **Rabi Hamiltonian** with the **Jaynes–Cummings model** by sweeping detuning and coupling strength to quantify counter-rotating effects and estimate error bounds.
- Studied **open quantum system dynamics** using Lindblad master equations and **Landau–Zener transitions** for non-adiabatic regimes.
- Implemented time-dependent Schrödinger evolution with external driving, computed transition amplitudes via **time-dependent perturbation theory**, and numerically reproduced **Rabi oscillations**.
- Constructed photonic qubit representations and simulated multi-mode optical dynamics to analyze elementary **quantum gate operations**.
- Identified regimes of **RWA failure** and built a reusable simulation framework for **coherent control**, decoherence, and photonic quantum architectures.

Synthesis and Discovery of Electronic Phase Transitions in Te-Rich $NbNi_2Te_2$ Crystals |Dr Zakir Hossain, IITK Course Project, Modern Physics Lab *(Sept'25 - Nov'25)*

- Synthesized layered Ni–Nb–Te single crystals and characterized their structure and transport properties to identify correlated electronic behavior in a **van der Waals telluride system**.
- Synthesized crystals using **chemical vapor transport** with a 1:2:2 Nb:Ni:Te starting ratio, iodine transport agent, and a 950°C to 850°C temperature gradient over 14 days, producing cleavable plate-like crystals.
- Determined a Te-rich composition ($Ni_{1.05}NbTe_{2.45}$) via **SEM–EDX** and confirmed strong c-axis preferred orientation using **X-ray diffraction** dominated by sharp (00ℓ) peaks.
- Measured temperature-dependent resistivity $\rho(T)$ from 300–4 K, verifying heating/cooling consistency to assess hysteresis and transition order.
- Observed **bad-metallic behavior** ($\rho(300\,\mathrm{K}) \approx 450\,\mu\Omega\cdot\mathrm{cm}$) with two anomalies: a **transition near 275 K** (no hysteresis) and a **resistivity minimum near 111.5 K**.

Geometric and Dynamical Analysis of Function Vector Representations in Transformer Models

Dr Jivnesh Sandhan, Graduate School of Informatics, Kyoto University *(Two manuscripts in preparation for ACL 2026 Industry Track)* | *(June'25 - Ongoing)*

- Modeled internal transformer representations as **trajectory dynamics in a high-dimensional state space**, revealing structured temporal evolution rather than static embedding behavior.
- Identified robust **power-law scaling** and systematic **directional drift** across GPT-2 layers ($R^2 = 0.86$–0.89), indicating constrained, non-random dynamics.
- Demonstrated that representation dynamics lie on a **low-dimensional manifold** (8–10 dominant modes, 96% variance), enabling accurate long-horizon forecasting using low-rank state-space predictors.
- Achieved **high-fidelity trajectory prediction** (cosine similarity up to 0.91) while reducing computational cost by **22–50%** relative to full-model evaluation.
- Diagnosed geometric failure modes in behavioral steering caused by a dominant **shared subspace** (83% alignment), collapsing functional separability.
- Developed a fast **orthogonal projection framework** that restores geometric independence, yielding up to **13.5× improvement in separability** and **18–22% bias reduction** at negligible $\mathcal{O}(nd)$ cost.

Course Presentations

Point Groups in 3D Crystal Structures *(Nov'25)*

Dr Subhomoy Haldar, IIT Kanpur

PHY312: Quantum Processes in Low Dim. Semiconductors

- Presented crystallographic point groups as symmetry operations governing anisotropy in optical, electrical, and mechanical properties of crystals, applying representation theory and character tables to identify orbital symmetries and degeneracies.
- Derived IR and Raman selection rules, explaining the role of inversion symmetry and the rule of mutual exclusion.
- Compared Si (O_h) and GaAs (T_d), and analyzed Rashba–Dresselhaus spin–orbit effects and symmetry breaking in monolayer MoS enabling spin–valley physics.

Variational Circuits and Quantum Machine Learning

Dr Washim Uddin Mondal, IIT Kanpur *(Mar'25)*

EE798V Project: Quantum Computing and Comms.

- Studied variational quantum circuits with parameterized gates for hybrid quantum–classical optimization, focusing on Variational Quantum Eigensolver (VQE) and quantum machine learning workflows.
- Analyzed quantum clustering algorithms (k-means, k-median, kNN); compared asymptotic runtimes with classical methods.
- Designed quantum pipelines for distance estimation via amplitude & angle embedding; implemented the swap-test circuit.
- Evaluated trade-offs b/w amplitude and angle embedding in terms of qubit cost, circuit depth, NISQ hardware feasibility.

Community Work

Programming Club |Coordinator *(April'25 - Ongoing)*

Secretary *(June'24 - April'25)*

- Led a two-level team of over 30 students to organize campus-wide hackathons, technical sessions, and project activities that increased student participation in programming and computational thinking.
- Mentored 20+ students on quantum algorithms and quantum machine learning using IBM Qiskit.
- Managed a budget of approximately INR 3 lakhs, coordinating vendors and ensuring smooth execution of large-scale events.
- Designed problem statements and evaluation frameworks for national and intra-campus hackathons.
- Collaborated with college administration to develop campus-level technical applications & improve computational services.